

Frontier AI Security Residency (FASR) Application

[ERA](#), in partnership with the [Oxford Hardware AI Governance lab](#) and the [Heron AI Security Initiative](#), is now accepting applications for our new [Frontier AI Security Residency](#): an 8-week, fully-funded residency for researchers & engineers working to build the security backbone for frontier AI systems.

Frontier AI is being built and deployed faster than it can be secured. Models are growing more capable month by month, and the infrastructure serving them expands with each data centre investment, presenting an ever-increasing attack surface. Through the **Frontier AI Security Residency**, we want to bring strong security talent to the hardest unsolved problems around frontier AI: protecting model weights, the infrastructure they run on, and the systems built on top of them.

The residency will feature two core focus areas, AlxCyber and AlxHardware, with 10-15 residents on each. Within the focus areas, we will also have an applied track and a research track:

Applied track projects will focus on building, red-teaming, or prototyping specific AI security tools and hardware.

Research track projects will focus on clarifying conceptual or strategic questions related to AI security and hardware

Alongside your track preferences, this application will also ask you to indicate which mentors you would be most interested in working with. Please have a look through our list of mentors before completing the application: <https://www.securefrontier.ai/mentors>

Notes on the application process (*please read*):

This form should take **45-60 minutes**. The form doesn't save partial progress so we recommend that you draft your answers separately, then copy them here.

We **encourage you to apply** even if you are unsure if you meet all the criteria, or if you feel you don't have the right background. We're particularly keen to hear from experienced cybersecurity & hardware security professionals, including but not limited to offensive researchers and red-teamers, systems engineers, cryptographers, PhD students and working professionals. This is a non-exhaustive list and we encourage anyone who is interested in this residency to fill out the form.

A note on AI tools. You're welcome to use AI tools or talk to others while preparing your application. In fact, we assume that many of you will, and we don't want to penalise it. What we're assessing is your judgement, your work, and your specifics. Generic answers score poorly regardless of how they were produced, and everything you claim is fair game for a live deep-dive at later stages of the application process. The best applications tend to be the ones that couldn't have been written by anyone but you.

Please note that while we can sponsor visas, you must ensure that you have at least one blank page available for the visa vignette in your passport, and that your passport is valid until June 2027 minimum. If this is not the case, we recommend starting the passport renewal process now, as it can take some time.

The deadline for applications is 2nd August 2026, 23:59 UTC.

If you have any questions, please read the FAQs on our website first at <https://www.securefrontier.ai/faq> and then contact us at hello@erafellowship.org.

1. Personal Information

What is your full name? *

Shiam Junior Chuttoo

Email *

[shiamjuniorchuttoo95@gmail.co
m](mailto:shiamjuniorchuttoo95@gmail.com)

Email-retyped *

[shiamjuniorchuttoo95@gmail.co
m](mailto:shiamjuniorchuttoo95@gmail.com)

Preferred first name: *

We will use this to refer to you throughout the interview process

Shiam

18 or over? *

Unfortunately, at this time, we cannot accept anyone under the age of 18.

☐

(Optional) Preferred pronouns

+

Of which country (or countries) are you a citizen? *

This information helps us to identify visa requirements. If needed, we will support successful applicants to secure a Visa for the duration of the Fellowship. Your response to this question does not affect the selection process in any way.

United Kingdom Mauritius +

The residency runs full-time, in person in Cambridge, UK, from Oct 12 - Dec 4, 2026. Can you commit to being present for the full programme? *

☐ Yes

☐ Yes, with caveats (specify below)

☐ No

2. Your Experience

LinkedIn *

If you don't have a LinkedIn, please put "N/A"

<https://www.linkedin.com/in/shiam-chuttoo-3aa7...>

CV/resume *



Shiam_Chuttoo_CV_Jul...

What is your current career level? *

Early career (0-5 years post-study)

What university did you attend for your undergraduate degree? *

Please put N/A if you didn't attend any university

Derby

Can you share a work sample which best represents your core speciality? *

Your application will be considered regardless, so don't worry if not. This could include e.g., a repo, write-up, paper, CVE or bug-bounty report, CTF profile, talk, tool, or product. If your best work is confidential, describe it at whatever level you can.

☐

Yes

☐

No

Clear selection

How will you share your work sample? *

☐

Link

☐

File upload

Clear selection

Link your work sample *

Please ensure that it is accessible to be viewed.

[https://github.com/Shiam1995/Ensemble-Model...](https://github.com/Shiam1995/Ensemble-Model-...)

Describe your role in this work *

What were your specific contributions - especially if this was created in a team?

This was my masters project, I was responsible for the ideation, I wanted a topic that was novel but one I was confident that I can do well in, enough familiarity that I could fall back on my skillset, but enough to experiment and try something new, I choose deepfake detection methods this was the time Sora (chatgpts video generator) first came out, I had to create a pipeline of to get the video and preprocess the video. I then had to work out what the previous method of deepfake were, what worked what didnt and how they would

be used, I went for the ensemble approach based on an idea from one of previous projects that different models handle different tasks well. I finished the job getting my distinction in the end.

(Optional) GitHub

<https://github.com/Shiam1995>

(Optional) Portfolio / Google Scholar

<https://www.youtube.com/@shiamjuniorchuttoo...>

3. Track & Mentor

The residency will feature two core focus areas, AlxCyber and AlxHardware, with 10-15 residents on each. Within the focus areas, we will also have an applied track and a research track:

Applied track projects will focus on building, red-teaming, or prototyping specific AI security tools and hardware verification prototypes.

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Please see [here](#) for our list of mentors and organisations.

Which focus area would you like to apply to? *

Choose the focus area which best fits your background and the problems you want to work on. Track descriptions are at securefrontier.ai/cyber and securefrontier.ai/hardware. You can select both if you're unsure.

☐ AlxHardware

☐ AlxCyber

Which track would you want to be on? *

You can select multiple.

☐ Applied

☐ Research

☐ Founding / Fieldbuilding

Please select up to 5 mentors that you would be interested in working with. *

Browse mentors and their research areas at securefrontier.ai/mentors. Indicating your top choice mentor here tells us who you'd most like to work with — it doesn't limit who can review your application, and there's no penalty for choosing a popular mentor.

Please note, we are still adding mentors to our website and not all appear on the application. This is only a selection; there will be more working on the programme than appear in this list.

Mengyuan Li ✕

Dominic Rizzo ✕

Jonathan Happel ✕

Ben Hagag ✕

Naci Cankaya ✕

+

Please rank the 5 mentors you selected, from most to least preferred, and explain your reasoning. *

Please note, your answer does not 'bind' you to a particular mentor.

Mengyuan Li :

I like Mengyuans work because I think trusted execution environments and an area which needs to be addressed.

I work in the nuclear sector, and what stops us using LLM agents in design work isn't capability, it's that we can't justify what they did or their reasoning, I think establishing the agents environment is a key step in

solving the trust problem.

Dominic Rizzo:

I like Dominics work because I think supply chain integrity is an important area to understand, and open source or verification of this is a key problem.

One of the problems in nuclear fusion is there are thousands of first of a kind parts, and many open tools are used(more software then hardware) but owning a domestic supply chain is key, but to own it we need to understand what we need to make and why.

Johnthan Happel:

I like Johnthan's work because I think it is easy (maybe not the best term) to design an idea but to bring it to market and start to understand the regulatory framework is something that will help me become a more complete researcher. I like the idea of 'turn informal trusted-compute ideas, including chip inspection and FPGA randomization, into a concrete threat model, taxonomy, and protocol sketch' into something more useful.

Ben Hagag:

Naci Cankaya :

If we can't match you with a mentor you've ranked, are you open to working with another mentor in your track?

Yes



4. Motivation & Reasoning

Which best describe your likely next step after the residency? *

Choose up to 2 options that you see as most likely for yourself.

- ☐ Apply for a Masters/PhD programme in frontier AI security / verification
- ☐ Apply for postdocs and/or professorships in frontier AI security / verification
- ☐ Apply for jobs at top AI security / verification research organisations
- ☐ Work in a government policy role (e.g. Congress)
- ☐ Work in a non-government policy role
- ☐ Found a start-up relevant to frontier AI security / verification
- ☐ Found a non-profit relevant to frontier AI security / verification
- ☐ Work in a sector unrelated to this field
- ☐ Keep my current job/role
- ☐ Other (specify)

Why is this residency the best use of your time for two months? What do you hope this residency will enable you to do afterwards? (max. 1200 characters) *

Please be as concrete as possible; we're most interested in how this residency fits your specific trajectory. Include any potential trade-offs or uncertainties.

My path wasn't planned: psychology, then an MSc in AI, now an EngD applying LLM agents to fusion plant design. It came together through circumstance more than design, but the result is a foundation broad enough that I can now go deep, and depth on one problem is the right move rather than another pass across many. Two months is also short enough to sit inside and augment my EngD rather than interrupt it.

I could read my way to a rough understanding of the space, but in my experience of eliciting stakeholder demands and requirements and recently going to hackthons and events people on the ground have different pain points and frameworks more gear to applied solutions.

The timing is the main thing. I'm early enough that whatever I take from this still gets to shape the thesis, rather than arriving after the work is done and I can't use it. If I did this at the end I'd finish with a clear idea of what I should have been building for the last three years. Doing it now means the rest of the PhD gets done by a better researcher, and that compounds in a way that two months of my own work never would. The honest uncertainty is whether this lands in time to matter. Verification for AI agents may simply not be ready to put in front of it within my doctorate. I think the direction is right regardless.

Your input must not exceed 1,200 characters

How do you foresee advanced AI affecting either 1) cybersecurity or 2) hardware-enabled verification over the next few years? (max. 1200 characters) *

5. References

Please provide at least two references. You are welcome to provide a third person as a back-up, but this is optional and will not be held against you if you don't.

We won't contact them without telling you first, and even then, we will only contact your references if you progress to our first interview stage.

Reference 1: Name *

Reference 1: Email *

Reference 1: Role & relationship to you *

Reference 2: Name *

Reference 2: Email *

Reference 2: Role & relationship to you *

[optional] Reference 3: Name

For example, as a back-up if you think one of your above references could be too busy to reply. Leave blank if no.

Additional comments

6. Additional Information

Please describe in detail how you heard about the Frontier AI Security Residency. Please be as specific as possible about which factors were most important in your decision to apply. *

(This will not affect your application in any way)

Write 1-3 sentences, e.g. "I was forwarded an email from [X person] via the Y newsletter. I then talked to [Z person] who encouraged me to apply."

I saw a link on linkedin, I am a big fan of the potential of AI and I think this is an important aspect of it, If I want AI to become useful to my parents and not just me who can fire up a terminal these question/problems need to get solved.

I consent to sharing my data with ERA and partner organisations, which will be processed in accordance with ERA's privacy policy. *

[ERA's privacy policy is linked here.](#)

☐

How, if at all, may we share your details with other programmes & hiring managers in this space? *

- ☐ Full share. Share my name, CV/Resume, email, and written answers to application questions on this form.
- ☐ Limited share. Share my name, CV/Resume, and email, but do not share my written answers.
- ☐ Name and email only; do not share my CV/Resume or other answers on this form.
- ☐ Do not share. Keep my information for ERA's internal use only.

Clear selection

Would you be happy to answer some additional demographic questions? *

We do not use these answers during the assessment process, but we track demographic data as part of our approach to monitoring diversity and inclusion at ERA.

Any information you provide will be kept entirely confidential and will never be traced back to you as an individual.

- ☐ Yes
- ☐ No

Clear selection

(Optional) What gender do you identify with?

- ☐ Female
- ☐ Male
- ☐ Non-binary
- ☐ I identify with a gender not listed above (specify in next question)
- ☐ Prefer not to say

Clear selection

(Optional) What is your ethnic group?

- ☐ American Indian, Alaska Native, or indigenous origin
- ☒ Asian, Asian American, or Asian British origin
- ☐ Black, Black British, African American, African Caribbean, or African origin
- ☐ Hispanic or Latin American origin
- ☐ Middle Eastern, North African, or Arab origin
- ☐ Native Hawaiian/Pacific Islander origin
- ☐ White or European origin
- ☐ Not listed above (please specify)

(Optional) Any further comments? If you have any questions, please send them to hello@erafellowship.org.

Do not submit passwords through this form. [Report malicious form](#)
